Chapter 10
Imaginaries of Education and Innovation in the European Union

Xavier Rambla

10.1 Introduction

The connections between geography, politics and knowledge have driven the most influential current trends in the fields of innovation and education policies. These connections are far from intuitive through the lens of Western scholarship, however. Although it has long been recognised that international borders and administrative units link geography and politics, the role of knowledge has been largely neglected.

Postcolonial scholarship and critical geography have opened a fruitful conversation on the connections between these three concepts (Mignolo, 2002; Moisio, 2019). This analysis builds on these to offer insight into innovation and education policies within the European Union. Significantly, current debates on these policy domains retrieve the geographical concept of region, expand the reach of politics beyond the nation-state, and unveil intricate links between knowledge and politics.

In this chapter I claim that the innovation and education policies of the European Union are currently undergoing two interrelated processes whereby complex configurations of policy actors focus on 'regions' as institutional units. On the one hand, these policy actors look for legitimation, engage in complex configurations of political relations, and select more and less important issues by means of policy instruments such as performance indicators, in ways that convey a particular mode of expert knowledge. On the other hand, the functions of policy design, implementation and evaluation migrate in different directions across the main geographical scales of governance; namely, localities and regions, member states and the EU itself. The first two sections of the chapter outline the main aspects of these two

X. Rambla (✉)
Department of Sociology, Universitat Autònoma de Barcelona, Barcelona, Catalunya, Spain
e-mail: xavier.rambla@uab.cat

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2022

M. Parreira do Amaral, C. Thompson (eds.), *Geopolitical Transformations in Higher Education*, Educational Governance Research 17,
https://doi.org/10.1007/978-3-030-94415-5_10

trends. Then, subsequent sections analyze notable implications for innovation and education EU policies.

The European Union (EU) presents a significant case for the study of the relationship between politics, knowledge and geography. The EU is a hybrid institution that mixes features of international organizations and national governments, where core decisions normally require intensive consultations involving the institutional 'triangle' defined by the European Council, the European Commission and the European Parliament. While the Council is similar to an international organization where the heads of member states sit, the Commission is more akin to a national government that deals with standard areas of policy responsibility (e.g. budget, innovation, social rights, economy, internal market). The Members of the European Parliament are elected every 4 years. The Council issues recommendations addressed to the governments of the member states themselves. The Council chooses the president of the Commission, and the member states choose the commissioners, but the Parliament must approve the whole team.

Since the 1950s, the Commission has asked experts to participate in workshops in which guidelines for action are devised and issued. The area of education runs a panel of regular workshops that have inspired the tenets of the recently approved European Education Area, which prioritises cooperation in lifelong learning, language training and digitalization. The contribution of experts to these workshops has led the statistical office, EUROSTAT, to measure a large variety of issues regarding education and innovation across the member states and in specific regions within the states. On the grounds of this collaboration between experts and policy-makers, the Commission has drawn an array of maps that classify member states and regions according to relevant indicators and core priorities.

10.2 Performance Indicators as a Policy Instrument

Since the Lisbon Summit in 2001, the EU has run 10-year strategic plans that aim to achieve key benchmarks at the level of the whole Union, the member states and the regions. The Innovation Union and Education and Training 2020 have been two of the most high profile examples of the last decade. Crucial to my argument is the observation that these plans construct national and regional maps from which any citizen can identify whether she lives in a region that leads innovation and provides sufficient education and training.

Le Galès (2016) distinguishes a set of policy instruments that combine techniques and power in variable ways. In this perspective, while the technical ingredient of these instruments is not reducible to the power ingredient, they are interrelated. Instruments such as legal and economic discourses, statistics, incentives and communications have their own rationality, but in practice are enacted through social relations between (more or less powerful) people with varying degrees of influence over others.

Performance indicators are a policy instrument that directly draws on scientific and technical legitimacy, and pushes policy actors towards competition (Le Galès, 2016). Indicators constitute rankings that indicate which practices decision-makers should adopt in order to improve their relative position with regard to other institutional or geographic units. While sometimes indicators measure the success of policy, quite often they signal a problem that had not been identified previously. Through interaction with other actors who have a stake in a given issue, policy-makers choose an instrument to monitor performance indicators in order to reflect on the strengths and weaknesses of the relevant alternatives. This instrument will greatly contribute to shaping the range of these alternatives, thus privileging some of them over others. In this sense, performance indicators become an objective reality that actors can neither invent from scratch nor fashion at will.

Table 10.1 identifies the main characteristics of performance indicators as policy instruments (Le Galès, 2016). Two of them operate by using expert knowledge in the midst of social interaction with other actors (Collins, 2018). The other two constitute social relations in which policy-makers face more opportunities of success if they pursue some objectives instead of other ones.

Experts construct knowledge in continuous interaction with other experts and policy makers. Some professionals become experts insofar as they sponsor certain recommendations that may help to prevent nuclear accidents, decide on social reforms, maintain macro-economic equilibria, tackle climate change, respond to the COVID-19 pandemic and solve many other collective problems. The ways in which experts on innovation and education gain influence is similar to those in a variety of other fields. Their reputation not only depends on their technical mastery but also on how the public perceives the success of these guides for action (Collins, 2018). Performance indicators convey this kind of expert knowledge by showcasing best practices in comparison with average or poor practices (Le Galès, 2016). In Europe, EUROSTAT ranks member states and regions according to their accomplishments in innovation and education.

Performance indicators convey 'theories of change' that highlight how policies activate certain mechanisms in a given context (Pawson, 2006). For some decades, the European Commission and the European Council have gained leverage in various policy areas on the grounds that they have contributed to better regulation by disseminating such theories or causal narratives (Radaelli et al., 2013). Since the 2000s, the literature on education and Europeanization has shown how the EU institutions were creating a European space on these grounds (Dale & Robertson, 2009).

Table 10.1 Crucial dimensions of performance indicators as policy instruments

Type of instrument	Performance indicators, standards and best practices
Legitimacy through interactive expertise	Visible accomplishment through regional indicators Adoption of common 'theories of change'
In-built political relations	Relations between governments and civil societies
Selectivity	Sidelined but relevant issues

Source: Author's elaboration drawing on Le Galès (2016), Collins (2018) and Jessop (2007)

Later on, while the initial educational European education space focused on higher education, the European Commission and the European Council have also fashioned similar spaces in the areas of school, vocational and adult education.

International organizations have promoted some of the best-known theories of change. For example, in the 1980s the World Bank took stock of the comparative rates of return of instructional levels in order to require indebted governments to target educational spending on primary education. The OECD has disseminated a theory of change for economic, innovation and education policies that strongly relies on claims of a virtuous circle that links investment in research, pedagogies that promote creativity and measures that enable a high proportion of young people to graduate in tertiary education. More recently, the Global Education Report has circulated the assumption that many Sustainable Development Goals are interconnected, thus highlighting mutual benefits between education and innovation as well as other goals regarding poverty, health, human communities, consumption and production and the natural environment. In the EU, the Education and Training 2020 assumes that reducing early school leaving and promoting innovation will bring about powerful synergies to enable smart, sustainable and inclusive growth.

Performance indicators indirectly fashion certain relations between governments and civil societies. Civil societies include the business community, private providers of public services, non-profits, social movements and other actors. Performance indicators transform all these actors into stakeholders with a shared objective who adjust their activities to new guidelines. Instead of passing binding pieces of legislation that fix the role of each type of stakeholder, through the use of performance indicators governments attempt to persuade all of them to simultaneously compete and cooperate in order to carry out an activity (Le Galès, 2016). The knowledge that the data convey is expected to eventually embed in social practices.

Finally, a significant aspect of policy instruments has to do with the order of priorities. Policy instruments 'select' relevant policy issues in the arena of public debate and simultaneously induce actors to sideline other issues. Jessop (2007, 217) has convincingly argued that a liberal economic order frames EU employment and economic policy so deeply that it inevitably leads decision-makers to focus on some issues at the expense of others.

10.3 The Geographical Scales of Europe

The European Union has started to use regional rankings at the same time as territorial disparities exacerbated and the power of decision-making shifted between different scales of governance. As far as disparities are concerned, research has documented that the geography of venture capital investment is strongly concentrated in some metropolitan areas and in certain neighbourhoods within these areas (Adler & Florida, 2017). A comparative analysis of learning in the workplace across the member states of the EU has also unveiled divided geographies. This analysis has observed how many workers have a margin of maneuver to make judgements on their own work. Although this margin is restricted to the level of top management in many countries, in Scandinavia it is widespread across the rank and file. Lundvall and Rasmussen (2016) have estimated an index of inequality in continuous learning at work based on the relative probability of managers to access discretionary learning compared to other employees. In contrast with Scandinavia, such inequality is exacerbated in Anglo-Saxon and South Western European countries. Thus, the challenges of adult skill formation seem to emerge from both the organization of labour and the institutional regimes of public policies, and the combination of these two components is quite different across countries and regions.

Regarding the scales of governance, the power of decision-making has shifted in very complex ways. For instance, since the sovereign debt crisis the EU has gained leverage on financial regulation and monetary policy, which have become an overwhelming problem for some member states. Simultaneously, the EU has asked member states to build partnerships with regional and local authorities in order to deal with employment, innovation and education, which are extremely sensitive to spatial variation. In addition, this functional change has become the context of new struggles for mobilization and representation as well as new designs of public policy (Keating, 2009, 22).

Diverse political forces have underpinned the importance of regions. In Belgium, Germany, Italy, Spain and the UK, specific constitutional designs and ulterior conflicts have endowed regional governments with more responsibilities. In France and some Central and Eastern European countries, the leading impulses have come from technocratic assumptions about the potential of regional governance as such. The strategic relevance of regions to apply for EU funding has also convinced some governments to proceed to regionalization within their countries. The growing protagonism of regions and cities has triggered contentions between those who believe that regional development should be taken out of politics and those who see regions as a privileged space for political mobilization (Keating, 2009, 41–3). Although sheer de- politization has seldom been successful, the role of sub-national authorities has been weakened by ex-ante conditions and top-down evaluations that have constrained the debate and the available alternatives (Sbaraglia, 2017).

The following sections explore the role of performance indicators as a policy instrument that derives some legitimacy from a widely shared ‘theory of change’. Roughly, this ‘theory’ assumes that innovation and education and training are two increasingly complex systems that should co-evolve and generate new synergies. The policy instrument also conveys certain expectations on the relations that policy actors must establish. Finally, the policy instrument enacts powerful patterns of selectivity that privilege some issues while downplaying others. In particular, I want to analyze how policy actors have used performance indicators to construct regions in the midst of social transformations that have exacerbated socio-economic divides between regions in Europe. This theme is very significant in the areas of education and innovation, which as indicated the EU has attempted to link throughout its territory. I will briefly outline the ambitious ideas that portray the potential of an Innovation Union and the economic contribution of the interface between education and training and employment policies (i.e., the Skills Agenda). Then, I will explore how the key dimensions of performance indicators as a policy instrument have enabled EU institutions to link these two discourses.

10.4 The Innovation and Education and Training Policies of the EU

The Innovation Union and the Skills Agenda are two flagship initiatives of the Europe 2020 Strategy that the European Commission and the European Council have deployed since 2010. The Strategy pursues interrelated and mutually reinforcing goals with respect to employment, research and development, climate change and energy, education and the reduction of poverty and social exclusion. A string of buzzwords condenses the whole approach into the idea of fostering smart, sustainable and inclusive growth (European Commission, 2010).

The Innovation Union has implications for both competitiveness and regional policy. The official standpoint insists on the need to invent new methods, products, approaches and practices that can cope with fiscal problems, changing demographic patterns and emerging challenges in global markets. At the same time, regions are supposed to host the eco-systems of innovation. Thus, this discourse presents regions as sites of economic activity and social interaction between heterogeneous stakeholders (European Union, 2016, 6; European Commission, n.d.).

The Skills Agenda draws on active labour market policies (ALMPs) as well as on the EU Education and Training Strategy. The Agenda recognizes the potential of regions to avail of mutually reinforcing actions in the areas of employment, education and innovation. The Agenda endeavors to increase employment rates and improve productivity scores. It also encompasses education and training, including the challenges of teaching and the popular perception of some programmes such as Vocational Education and Training (VET). The informal occasions for learning in many sites are also a crucial component of the Agenda (European Commission, 2016: 2–3).

A series of Council Recommendations have translated these general principles into the concrete terms of education. These Recommendations have transcended the traditional tight coupling of education with services delivered to minors. Thus, the European Council (2011b) has called for the due consideration of adult education as a fundamental component of lifelong learning. Furthermore, the European Council (2011a) has blurred the distinction between compulsory and post-compulsory education in three significant ways. First, it has emphasized the responsibility of any educational authority for preventing early school leaving from primary education onwards. Second, it has underpinned intervention on the population perceived to be at risk, mostly in lower secondary education. Third, it has encouraged measures to compensate for the consequences of dropping out without basic academic credentials. Overall, this compensation conveys a new understanding of adult education and lifelong learning.

The European Council (2013) Recommendation on the Youth Guarantee Scheme has committed the domains of employment and youth work to deliver education and training to young adults who have already finished their compulsory education. The scheme seeks to ensure that any youth has the opportunity to engage in employment, education, training or apprenticeships within 4 months of leaving school or finishing a job.

Crucial to my analysis is an observation on the governance of the Innovation Union and the Skills Agenda: By blurring previously established boundaries between policy sectors, the interaction of Brussels-based institutions with national and sub-national governments posits a (perhaps not fully acknowledged) array of coordination problems to the actors involved. While Sbaraglia (2017) notes that regional authorities are exposed to new constraints due to increased surveillance, they also face growing problems of coordination that complicate their work as well as that of national and European authorities.

10.4.1 Accomplishing Benchmarks

The Innovation Union and the Skills Agenda have formed part of the vision for the European Union since its inception. In the 1990s, the Treaties that constituted the Union as such also sketched the legal scaffolding of these initiatives. In 2001, the Lisbon Agenda became the first attempt to translate those principles into reality. In 2010, the Europe 2020 Strategy simply followed suit. Over time, statistics have portrayed regions as a privileged site for monitoring the progress of those ideas.

Significantly, EUROSTAT has complemented the member states' Innovation Scoreboard with a Regional Innovation Scoreboard. This tool reflects a very particular accomplishment; namely, their capacity to build an 'innovation eco-system' that aligns education and training with investment, innovation activity and employment in each NUTS2[1] region (Hollanders & Es-Sadki, 2017).

In a similar vein, the educational regional indicators attempt to capture the 'education and training system' of the region, including an estimation of the participation rate of the whole population regardless of age. The indicators are also sensitive to diverse life courses insofar as they take early school leaving into account and estimate employment rates according to 'years since completion of higher level of education' (EUROSTAT, n.d.).

Thus, EUROSTAT draws a series of maps that rank member states and regions according to their capacity to innovate and their capacity to prepare both the young and the adult population. These maps classify the regions according to their score on continuous variables that may be measured across the Union by drawing on common statistical sources. It is quite clear, however, that the resulting indicators can capture neither all types of innovation nor the whole array of life courses. They can

[1] EUROSTAT uses the French acronym of *Nomenclature des unités territoriales statistiques.*

only report on numbers of technical and scientific staff, amounts of resources invested in R + D, and the instructional status of people at a given age.

10.4.2 Sharing 'Theories of Change'

The labels Innovation Union and Skills Agenda convey 'theories of change'. The Innovation Union represents a joint endeavor to build an 'innovation eco-system'. *Inter alia*, such a system should embrace research and market infrastructures within the EU as well as partnerships and coherent strategies that link EU and non-EU actors. In the official view, a common strategy in all these areas will foster the strengths of innovation in the EU. The experts who evaluated this initiative in 2016 considered that the main commitments should reinforce consistency at the level of the EU, for instance, by reducing fragmentation, coordinating procurement, pooling forces and designing external leverage carefully. Therefore, when the EU published the corresponding report, it recognized that the main potential lay in creating a coherent institutional whole (European Union, 2015).

The official expectations of the Skills Agenda also rely on building a system, that is to say, an institutional whole that can transcend traditional policy sectors such as employment, school education and VET. Actually, the key points of leverage operate in the midst of these sectors: skills formation requires both education and training and active labour market policies. Visible skills facilitate transitions between educational levels and programmes as well as enable individuals to find meaningful jobs. Skills intelligence may become an instrument for schools, training providers and employers (European Commission, 2016, 3).

For the last decade, the Council has envisioned an education and training system that fully renovates previously established sub-sectors of education. Thus, instead of compensatory, urban or priority education targeted to at-risk (often, secondary education) students, this view endorses a preventative and direct approach to reducing early school leaving by following the transitions within the school system and catering to the needs of young people above the school leaving age. In a similar vein of renewal, the Council wishes to expand the action of adult education beyond the low-skilled groups of a middle-aged population in order to deliver high-quality learning to everybody, regardless of class, gender, ethnic or other social divides (European Council, 2011a, C191/2; European Council, 2011b, C372/3).

Similarly, the Youth Guarantee Scheme has proposed to articulate education, job search and traineeships on the grounds of a wider concept of innovation, education, training and employment policies in the European Union. Instead of an ad-hoc solution, the list of options that member states and sub-national authorities have been advised to guarantee to young people forms a key component of this system that associates the Innovation Union with the Skills Agenda (European Council, 2013, C120/3).

In sum, the innovation system is expected to co-evolve with the renovated and expanded version of the traditional education and training system. The official documents clearly state that this mutual interaction is likely to enact at least three synergies that will greatly contribute to the goals of the Europe 2020 Strategy.

First, a key commitment calls for all member states to deliver new opportunities by articulating their own innovation and education policies. This commitment should deliver for 'low-skilled adults in Europe'. It should also curb any 'process whereby certain individuals are pushed to the edge of society'. Moreover, this approach should contribute 'to meeting the Europe 2020 goals of reducing early leaving from education and training to below 10%'.

> To improve the employment opportunities of low-skilled adults in Europe, Member States should put in place pathways for upskilling via a Skills Guarantee established in co-operation with social partners and education and training providers, as well as local, regional and national authorities. Upskilling should be open to people both in-work and out of work. Low-skilled adults should be helped to improve their literacy, numeracy and digital skills and – where possible – develop a wider set of skills leading to an upper secondary education qualification or equivalent (European Commission, 2016).

> Adult learning can make a significant contribution to meeting the Europe 2020 goals of reducing early leaving from education and training to below 10%. Particular attention should accordingly be paid to improving provision for the high number of low-skilled Europeans targeted in Europe 2020. At the same time, the substantial contribution which adult learning can make to economic development — by strengthening productivity, competitiveness, creativity, innovation and entrepreneurship — should be recognised and supported (European council, 2011b: 3).

Second, official documents expect that awareness be raised of skills shortages and skills mismatches among 'national, regional and local governments, regional grass-roots organizations and social partners but also actors at European level'. There are numerous references to the requirement that 'no one must be left behind' and that ask governments to 'reduce the share of 15-year-olds only achieving low levels of reading, mathematics and science to less than 15% by 2020'.

The digital transformation of society, the economy and industries across all sectors will have a far-reaching impact for Europe and its citizens. It presents a major opportunity for Europe but is also accompanied by a number of challenges. In particular, Europe must ensure that its citizens and its labour force have the appropriate digital skills to live and work in the new digital era. No one must be left behind (…) [The Commission recommends member states to] Harness and include those networks and actors who are providing the solutions on the ground, at national and local level; national, regional and local governments, regional grass-roots organizations and social partners but also actors at European level such as the Digital Champions (European Commission, 2018).

Third, the official messages align social cohesion with territorial cohesion. Thus, this discourse encourages member states to 'promote a harmonious economic, social and territorial development of the Union as a whole' by means of cooperation between regions. These arguments often appeal to 'social innovation' and 'social partners' (European Union, 2016).

10.4.3 Networking Levels of Government and Local Civil Societies

The discourse of the Innovation Union assumes that certain policy actors are the main targets of these recommendations. Both the Council and the Commission repeatedly mention 'the public sector, business, academia, finance' and other sectors as key stakeholders (European Commission, 2016, 7; European Union, 2016, 68).

To this end, the Innovation Union introduced a more strategic and broad approach to innovation by including actions that aimed to tackle both the supply and demand side elements of the innovation eco-system: the public sector, businesses, academia and finance. It equally assigned responsibilities and actions among the actors with the ability to shape the framework conditions for innovation, from the European Commission to Member States and Regional Governments, as well as other relevant stakeholders (European Union, 2015).

The employment and education policies that implement the Skills Agenda are also addressed to these networks of stakeholders. For instance, the European Social Fund (ESF) is open to social partners. The ESF supports training and active labor market policies that greatly contribute to underpin the 'education and training system' (European Social Fund, n.d.). As far as education is concerned, tackling early school leaving is not the exclusive responsibility of schools; on the contrary, "all relevant stakeholders" are invited to have a say (European Union, 2011b, C191/2-3). Further, the Council has recommended that member states 'develop partnerships' with a variety of stakeholders in order to implement the Youth Guarantee Scheme (European Union, 2013, C120/4). The Council has also reminded member states of the importance of civil society in the domain of adult education (European Union, 2011b, C372/4).

> ACCORDINGLY [THE COUNCIL] INVITES THE MEMBER STATES TO Ensure effective liaison with the relevant ministries and stakeholders, the social partners, businesses, relevant non-governmental organisations and civil society organisations, with a view to improving coherence between policies on adult learning and broader socio-economic policies (European Union, 2011b: C372/4).

At this point, a selection of findings from qualitative research projects will help to explore the reception of these policy messages in some regions. Project YOUNG_ADULLLT interviewed 168 experts on lifelong learning policies in Upper Austria, Vienna (Austria), Blavoevgrad, Plovdiv (Bulgaria), Bremen, Frankfurt (Germany), Girona, Málaga (Spain), Kainnu, Southwest Finland (Finland), Istria, Zagreb (Croatia), Genoa, Milano (Italy), Litoral Alentejano, Vale do Ave (Portugal), Aberdeen and Glasgow (United Kingdom). This data showed that variable but widespread configurations of bureaucratic and network governance implement these policies across the EU. Market governance does not seem to be so influential as in the United States, where private training providers are active almost everywhere (Rambla et al., 2018).

YOUNG_ADULLLT also showed the relevance of official ‘theories of change’ for the street-level educators professionals who directly interact with the beneficiaries of lifelong learning programmes. In the interviews, not only were many of them aware of the official discourse of the EU, but also, in some countries, they had developed a systematic theory of change. These common frameworks greatly contributed to articulate bureaucratic services with the professional networks of non-profits and the personal networks of the beneficiaries on the ground.

In a number of places, pre-existing initiatives were referred to. In Austria and Germany, the theory of change was expressed in terms of apprenticeship systems, whereas in Finland ‘Public-Private-People-Partnerships’ were referred to, and the ‘Employment Pipeline’ in Scotland. In each case, the interviewees consistently referred to the tenets and theses of the corresponding theories. In contrast, in places where the discourse of ‘education and training systems’ had more recently been introduced through EU documents, professionals seemed to struggle to overcome negative stereotypes of young people who applied for either training or welfare (Rambla et al., 2018).

A comparative discussion of Liguria (Italy) and Catalonia (Spain) sheds light on the role of local politics in reducing early school leaving (Bartolini, 2018; Tarabini et al., 2017). In Italy and Spain the national and the regional governments have not been particularly sensitive to the preventive and compensatory approach that the Council and the Commission have promoted. In both cases, however, some second-chance schools have been particularly innovative in elaborating wholesome ‘theories of change’ in relation to compensating for early school leaving. These schools have explicitly evolved from a remedial, last-chance understanding of educational services for early school leavers to educational, ambitious and sophisticated visions of these services. In Catalonia, second-chance schools have generated a new form of political relations. Here, some non-profits have created the model independently and then asked for public support. Municipalities have then designed a public policy on these grounds.

In sum, the EU has emphasized invitations to civil society to join its endeavor to build an ‘innovation eco- system’ as well as a wide-ranging ‘education and training system’. Although it is too early to draw general conclusions, a significant and varied array of partners is playing an important role in EU regions. Civil society networks engage in relationships with public employment services in all sites involved in the research. In Germany and the neighboring countries, these networks include the traditional partners of neocorporatism, i.e., government, employers’ associations and trade unions. Although the networks are much weaker in Southern (both Eastern and Western) Europe, some spurs of collaboration between local non-profits, employment services and education and training systems are emerging. Broadly speaking, local and regional professionals are taking the official ‘theories of change’ into account, although the habit of using these narratives of public policies is not common everywhere. Significantly, in Italy and Spain a few non-profits and even a few schools have designed innovative ‘theories of change’ that cater to the needs of early school leavers.

10.4.4 Homogenizing Innovation and Work-Centred Lifelong Learning in All Regions

Regional policies and civil societies enact mechanisms of selectivity that override a whole array of sensitive themes for many regions and localities. Selectivity consists of privileging some political alternatives at the expense of others. Thus, the policy actors who endorse certain alternatives are in a much more comfortable position than the ones who endorse other alternatives (Jessop, 2007). The acknowledgement of regional circumstances and the scope of lifelong learning programmes seem to be particularly exposed to selectivity.

For example, the ideal of building 'innovation eco- systems' posits a first illustration. Significantly, the Commission addresses all the regions in the 28 member states of the European Union. However, the Regional Innovation Scoreboard has clearly shown that these regions are quite dissimilar. In terms of the measures of innovation, while Scandinavian, many German-speaking and Western Atlantic regions are very innovative, other regions are very modest innovators (Hollanders & Es-Sadki, 2017). The general discourse, however, assumes that all these disparate cases will fit into the same type of 'systems', thus overlooking a huge variety of particular circumstances.

Lifelong learning policies posit a second instance of homogenization. In general, lifelong learning includes both objectives related to personal development and objectives related to employment and productivity. According to the findings of YOUNG_ADULLLT, the humanistic strand was noticeable in Finland, where some professionals told the interviewers that job centres and educational programmes were open to all people who experienced diverse biographical circumstances. However, lifelong learning had a different meaning in all the other countries. The professional interviewees associated this label with active labor market policies that were mandated to find jobs for beneficiaries in a short period. Additionally, although the respondents acknowledged that learning was necessary along the whole life course, most of them took for granted that these policies were basically an instrument to curb the number of youth who were unemployed (Rambla et al., 2018).

Ultimately, performance indicators not only measure but also define 'innovation eco- systems' and 'lifelong learning'. While the social agents that operate in different regions may develop very different types of innovation, they cannot showcase their 'eco-system' unless they measure according to specific metrics, e.g. R + D spending and patents. Similarly, while people learn throughout their life in the course of carrying out human activities such as employment, family responsibilities, political participation and leisure, their biographical experiences are not relevant to measures of learning and innovation unless they are directly related to employment. Performance indicators enact selectivity to the extent that they narrow down the diversity of the eco-systems and the scope of meaningful transitions during the life course of people.

10.5 Implications of EU Innovation and Education and Training Policies for Regions

In the European Union, complex configurations of politics, geography and expert knowledge have fashioned the imaginaries of innovation and education and training. The main EU institutions interact not only with member states but also with regions and cities. A wide range of political processes is enriching the landscape of regional policies, but by applying performance indicators to innovation and education and training, the EU institutions have produced their own knowledge on the main challenges that member states, regions and localities must face.

Table 10.2 outlines in which ways the EU construes an image of regions through innovation policies as well as through education and training policies. Drawing on the dimensions of performance indicators as a policy instrument (see Table 10.1), the table highlights how policy instruments deal with regions. Each row summarizes the observations developed in the previous four sections.

First, accomplishment is the most explicit regional dimension of the policy instrument. Since EUROSTAT estimates regional indicators of both innovation and education, any policy-maker can easily consult a map that classifies her region according to a variety of rankings that distinguish active and modest innovators as well as more or less effective systems of education and training.

Second, the intellectual underpinning of these policies lies in ‘theories of change’. The endeavors to make an ‘innovation eco-system’ and a wide-ranging ‘education and training system’ entail several beliefs on the possible causal effects of the initiatives that constitute these systems. In both cases, regions become the anchor of the systems.

Third, the official expectation of crafting appropriate political relations affects both innovation and education and training. The idea is that a diverse variety of

Table 10.2 Implications of EU innovation and education and training policies for regions

Performance indicators as policy instrument	Innovation	Education and training
Accomplishment of benchmarks	Regional innovation scores	Regional (NUTS2) statistics
‘Theories of change’	Innovation eco-systems rooted in regions	Education and training systems (schools, action on early leaving, adult education, YGS) rooted in regions
Political relations	Civil societies to build networks gathering national school systems, regional and local educational authorities, private training providers, VET institutions, ALMPs, employers, business, finance, unions and non-profits	
Selectivity	Overlooks regional polarisation	Except for Finland, ‘employment-first’ approaches prevail

stakeholders collaborate with authorities to foster those two types of system. Some partial evidence suggests that the articulation of bureaucratic and network governance at the local level is generating new political relations in the domain of education and training.

Fourth, selectivity eventually impinges on the discursive construction of regions by obfuscating the relevance of some issues. It is telling that the European institutions assume that 'innovation eco-systems' flourish in technological hubs as well as in settings that are not so well connected with the main trends of economic globalization. Similarly, by privileging job search over many other elements of education, lifelong learning policies also tend to depict regions as labour markets regardless of many other components of geography.

In brief, the geopolitics of knowledge illuminates key processes that take place within the European Union. Politics intermingles with knowledge insofar as decision-makers want the public sector, business and individuals to come up with innovative practices that strengthen the economy. Politics and knowledge are also the basis of education and training policies. The Innovation Union and the EU Skills Agenda understand that the regions of the EU will avail of unprecedented synergies if policy-makers are able to build encompassing innovation systems and education and training systems. The image of synergies between these two types of system has become a taken-for-granted portrait of regions in the official discourse. However, it is also notable that this portrait privileges particular aspects at the expense of excluding others. While the mainstream message emphasizes the expectation to trigger a virtuous circle of innovation and education, the Union sidelines concerns with regional heterogeneity and lifelong learning for human empowerment.

10.6 Conclusions

The European Union has elaborated quite sophisticated imaginaries of education and innovation by means of policy instruments that use expert knowledge to draw certain geographical images of the member states and the regions within these states. Remarkably, these maps have proliferated at the same time as the locus of decision-making has shifted between geographical scales in very complex ways. In doing so, the top EU institutions (i.e. Council, Commission and Parliament) have utilized performance indicators that measure the outcomes of education and training and estimate the innovative potential of regions.

In order to accomplish measurable benchmarks that locate their areas of responsibility in comparative rankings, the European Commission and the European Council have actively encouraged national and sub-national decision-makers to adopt best practices in the fields of education and innovation. These recommendations normally assume theories of change relating to imagined potential synergies between innovation eco-systems and education and training systems, which are expected to open new opportunities for citizens as well as to detect mismatches between skills and labor markets that should be urgently addressed.

Performance indicators have also asked governments, employers, educational institutions, technical centres and other stakeholders to collaborate with as the widest possible networks s that gather varied policy actors in their areas and their regional context. When focusing on benchmarks and relying on theories of synergy, decision-makers have not only woven these large networks but also naturalized a common order of priorities. Significantly, these priorities have privileged linear concepts of R + D and employment-centred understandings of lifelong learning.

In a nutshell, the EU has used education, science and innovation to design a wide-ranging variety of policies. These three components have contributed to shape the mainstream imaginaries of the knowledge society. In all policy areas, including innovation as well as innovation and training, decision-makers meet experts who sponsor certain policy instruments. Some strands of expert knowledge are built into these instruments. Insofar as the EU regulates its innovation and education and training policies by means of performance indicators, the EU member states and regions become the units of official rankings that classify the quality of professional and institutional practices. This political and technical operation has implications for political relations at all levels of governance as well as for the order of priorities. It is important to spell out these implications in order to find out the extremely significant linkages between politics, knowledge and space. Moreover, research on these implications also provides empirical evidence on the margin for democratic deliberation and the predicaments that consistent and feasible proposals to broaden up the scope of democratic deliberation would likely face.FundingThis analysis is an outcome of the projects YOUNG_ADULLLT and EDUPOST16. YOUNG_ADULLLT has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 693167. EDUPOST16 has received funding from the Government of Spain R&D programme under the grant agreement CSO2016-800004P.

References

Adler, P., & Florida, R. (2017). *The city and innovation* (Working paper series Martin prosperity research, 2017-MPIWP) (pp. 1–37).

Bartolini, M. (2018). *La dispersione scolastica visibile e invisible. Rappresentazioni, significati e teorie del cambiamento di un fenomeno multiforme*. Università degli Studi di Genova (PhD Dissertation).

Collins, H. M. (2018). Studies of expertise and experience. *Topoi, 37*, 67–77.

Dale, R., & Robertson, S. (2009). *Globalisation and Europeanisation in education*. Symposium Books.

European Commission. (2010). EUROPE 2020 A strategy for smart, sustainable and inclusive growth. In *Communication from the Commission, COM(2010)* (pp. 1–35). European Commission.

European Commission. (2016). A new skills agenda for Europe. Working together to strengthen human capital, employability and competitiveness. In *Communication from the Commission to the European Parliament, the Council, the European Social and Economic Committee and the Committee of the Regions, COM(2016)* (pp. 1–18). European Commission.

European Commission. (2018). *Digital skills and jobs coalition action plan for 2018 and beyond.* Retrieved from https://www.eciia.eu/2018

European Commission. (n.d.). *European regional and urban development*. Retrieved May 15, 2019a, from https://ec.europa.eu/info/eu-regional-and-urban-development_en

European Council. (2011a). Council recommendation of 28 June 2011 on policies to reduce early school leaving. *Official Journal- European Union Information and Notices C, 54*(191), 1–6.

European Council. (2011b). Council resolution on a renewed European agenda for adult learning. *Official Journal- European Union Information and Notices C, 54*(372), 1–6.

European Council. (2013). Council recommendation of 22 April 2013 on establishing a Youth Guarantee. *Official Journal – European Union Information and Notices C, 56*(120), 1–6.

European Social Fund. (n.d.). *How the ESF works*. Retrieved May 15, 2019b, from http://ec.europa.eu/esf/

European Union. (2015). *State of the innovation union 2015*. Publications Office of the European Union.

European Union. (2016). *Smarter, greener, more inclusive? Indicators to support the Europe 2020 strategy*. Publications Office of the European Union.

EUROSTAT. (n.d.). *Database by themes: General and regional statistics*. Retrieved May 15, 2019, from https://ec.europa.eu

Hollanders, H., & Es-Sadki, N. (2017). *Regional innovation scoreboard 2017. Regional Report*. EUROSTAT.

Jessop, B. (2007). State power: A strategic-relational approach. .

Keating, M. (2009). Rescaling Europe. *Perspectives on European Politics and Society, 10*(1), 34–50.

Le Galès, P. (2016). Performance measurement as a policy instrument. *Policy Studies, 37*(6), 508–520.

Lundvall, B.-A., & Rasmussen, P. (2016). Challenges for adult skill formation in the globalising learning economy – A European perspective. *International Journal of Lifelong Education, 35*(4), 448–464. https://doi.org/10.1080/02601370.2016.1203364

Mignolo, W. (2002). The geopolitics of knowledge and the colonial difference. *The South Atlantic Quarterly, 101*(1), 57–96.

Moisio, S. (2019). Re-thinking geoeconomics: Towards a political geography of economic geographies. *Geography Compass, 13*, e12466.

Pawson, R. (2006). *Evidence-based policy. A realist perspective*. Sage.

Radaelli, C., Dunlop, C., & Fritsch, O. (2013). Narrating impact assessment in the European Union. *European Political Science, 12*, 500–521.

Rambla, X., Kovacheva, S., Walther, A., Verlage, Th., & Jacovkis, J. (2018). *International qualitative analysis report*. YOUNG_ADULLLT working paper. Retrieved from http://www.young-adulllt.eu

Sbaraglia, F. (2017). Profession: Europe? La gestion infranationale des affaires européennes. *Politique Européene, 57*(3), 8–20.

Tarabini, A., Jacovkis, J., & Montes, A. (2017). *Los factores de la exclusión educativa en España: Mecanismos, perfiles y espacios de intervención*. UNICEF.